NATIONAL AGRICULTURAL INTELLIGENCE ARCHITECTURE

FarmPulse Malawi

System Architecture, Mathematical Formulations, Model Specifications & Data Protocols

TARGET PLATFORM	GEOGRAPHIC SCOPE	TARGET BENEFICIARIES	INSTITUTIONAL VERSION
Django 5.1 / Python 3.12	Malawi (National & District Level)	Smallholders, AEDOs, Cooperatives	Release 2.5 (November 2026)

1. Executive Summary & Core Mission

FarmPulse Malawi is an integrated computational agriculture platform designed to de-risk smallholder farming enterprises, optimize land allocation, provide auditable input cost ledgers, and deliver localized climate-smart agro-economic advisories.

The platform eliminates emotional or arbitrary cropping decisions by bridging three critical data layers into a unified mathematical optimization framework:

- **Agro-Ecological Suitability:** Historical and live meteorological telemetry from DCCMS and agro-ecological zoning.
- **Spatial Market Intelligence:** Real-time wholesale and retail market prices across key trading hubs (AMIS, HDX/WFP).
- **Input Cost Verifiability:** Transparent itemized budgeting matching certified seed, basal fertilizer, top-dressing, labor, and transport to farmer liquidity.

2. System Architectural Framework

MODULAR COMPONENT ARCHITECTURE

Module / App	Primary Responsibility	Key Database Models	External Data Interfaces
farms	Farm spatial holding, soil classification, parcel records, seasonal history.	`Crop`, `Farm`, `Season`, `Planting`, `SeasonReview`	Farmer GPS coordinates, AEDO field audits.
climate	Agro-ecological zoning, weather station telemetry, rainfall onset, drought tracking.	`District`, `WeatherStation`, `DailyWeatherObservation`, `WeatherForecast`	DCCMS Meteorological Bulletins, CHIRPS/ERA5 precipitation models.
market	Commodity wholesale/retail rates, transport spatial matrices, price forecasting.	`MarketHub`, `DailyMarketPrice`, `MarketPriceForecast`, `TransportCost`	Ministry of Agriculture AMIS, WFP VAM price streams.
planner	Crop portfolio linear solver, input cost ledger, stress-testing risk arena.	`Plan`, `PlanItem`, `PlanAssumption`, `InputCostBenchmark`, `SeasonalInputCost`	HDX / WFP Malawi food and input price feeds.
innovation	Earth observation parcel telemetry, NDVI vegetation indices, parametric drought index.	`SatelliteParcel`, `ParametricInsurancePolicy`	Sentinel-2 MSI surface reflectance, thermal drought triggers.
core	Institutional UI tokens, user authentication, bilingual Chichewa/English translation engine.	`UserProfile`	Community radio audio distribution, SMS/Audio endpoints.

3. Database Models & Schema Specifications

3.1. Farm & Cropping Models (`farms`)

Model	Primary Fields	Field Types & Rules	Operational Meaning
Crop	name, slug, category, typical_yield_kg_per_ha, seed_rate_kg_per_ha, growing_days, water_req	CharField, SlugField, DecimalField, IntegerField	National crop directory covering staple, legume, tuber, and cash crops in Malawi.
Farm	owner, name, district, total_land_ha, soil_type, has_irrigation, latitude, longitude	ForeignKey(User), CharField, DecimalField, BooleanField	Represents the physical farm holding, geo-referenced to a Malawian district.
Planting	farm, crop, season, area_ha, planting_date, harvest_date, actual_yield_kg, total_expenses_mwk	ForeignKey(Farm, Crop, Season), DecimalField, DateField	Historical and active production instance recording real agronomic performance.
SeasonReview	farm, season, rating, challenges_encountered, lessons_learned, net_profit_mwk	ForeignKey(Farm, Season), IntegerField, TextField, DecimalField	End-of-season post-mortem enabling learning loops across farming years.

3.2. Meteorological Models (`climate`)

Model	Primary Fields	Field Types & Rules	Operational Meaning
District	name, region, agro_ecological_zone, normal_annual_rainfall_mm	CharField, DecimalField	Aggregates climate baselines across Northern, Central, and Southern regions.
WeatherStation	station_code, name, district, latitude, longitude, elevation_m, is_active	CharField, ForeignKey(District), DecimalField, BooleanField	Represents official DCCMS or community automatic weather observation stations.
DailyWeatherObservation	station, date, rainfall_mm, min_temp_c, max_temp_c, humidity_pct	ForeignKey(WeatherStation), DateField, DecimalField	Daily physical readings driving the dry spell and rainfall onset algorithms.
WeatherForecast	station, forecast_date, rainfall_prob, expected_rainfall_mm, advisory_code	ForeignKey(WeatherStation), DateField, DecimalField, CharField	7-day to 14-day operational forecasts triggering seasonal planting advisories.

3.3. Market & Economic Models (`market`)

Model	Primary Fields	Field Types & Rules	Operational Meaning
MarketHub	hub_code, name, district, market_type, latitude, longitude	CharField, ForeignKey(District), DecimalField	Wholesale/retail trading hubs (e.g. Chinkhoma, Lunzu, Lizulu, Mzuzu, Lilongwe).
DailyMarketPrice	hub, crop, date, wholesale_price_mwk_kg, retail_price_mwk_kg, volume_grade	ForeignKey(MarketHub, Crop), DateField, DecimalField	Tracks daily price movements in Malawian Kwacha per kilogram.
TransportCost	origin_district, destination_hub, distance_km, rate_mwk_per_ton_km	CharField, ForeignKey(MarketHub), DecimalField	Spatial freight cost model calculating farm-gate net revenue after transport haulage.

3.4. Decision Engine & Input Cost Ledger Models (`planner`)

Model	Primary Fields	Field Types & Rules	Operational Meaning
InputCostBenchmark	crop, input_category, description, benchmark_price_mwk_per_ha, season_year, data_source, last_synced_at	ForeignKey(Crop), CharField (6 choices), DecimalField, DateTimeField	Official seasonal benchmark prices for Seed, Basal Fert, Top Dress, Labour, Herbicide, Transport.
SeasonalInputCost	farm, crop, season_year, seed_cost_mwk_per_ha, basal_fert_cost, top_dress_cost, labour_cost, herbicide_cost, transport_cost	ForeignKey(Farm, Crop), PositiveSmallIntegerField, DecimalField (nullable)	Farmer-adjusted actual quotes overriding benchmarks to reflect real village agro-dealer prices.
Plan	farm, season, total_land_ha, total_budget_mwk, optimization_goal, expected_revenue, expected_profit, risk_score	ForeignKey(Farm, Season), DecimalField, CharField	Optimized crop portfolio blueprint balancing land, capital, labor, and risk.
PlanItem	plan, crop, allocated_ha, expected_yield_tons, expected_revenue, input_cost, labour_cost, net_margin	ForeignKey(Plan, Crop), DecimalField	Detailed per-crop allocation breakdown resulting from the optimization solver.
PlanAssumption	plan, parameter_name, value_used, source_note	ForeignKey(Plan), CharField, DecimalField	Immutable audit snapshot freezing input prices and yield assumptions at generation time.

4. Mathematical Formulations & Optimization Algorithms

4.1. Linear Programming Crop Allocation Solver

The planner determines the optimal land vector $\mathbf{x} = [x_1, x_2, \dots, x_n]^T$ where x_i is the area (in hectares) allocated to crop $i \in C$.

> **Decision Variables:** $x_i \geq 0$ (Hectares allocated to Crop i)
>
> **1. Maximum Profit Objective:**
> $\text{Maximize } Z = \sum [(p_i \cdot y_i - c_i) \cdot x_i]$
>
> **2. Risk-Balanced Objective:**
> $\text{Maximize } Z_{balanced} = \sum [(p_i \cdot y_i - c_i) \cdot x_i] - \lambda \cdot \sum [r_i \cdot c_i \cdot x_i]$
> where λ is the farmer risk aversion coefficient ($\lambda = 0.35$ by default)
> r_i is the risk vulnerability index of crop i (0.0 = low risk, 1.0 = high risk)
>
> **3. Operational Constraints:**
> - Total Land Constraint: $\sum x_i \leq \text{Total_Arable_Land_ha}$
> - Cash Liquidity Constraint: $\sum (c_i \cdot x_i) \leq \text{Total_Available_Budget_MWK}$
> - Household Food Security: $x_{maize} \cdot y_{maize} \geq (\text{Household_Size} \cdot 220 \text{ kg})$
> - Agronomic Rotation Cap: $x_i \leq \mu_i \cdot \text{Total_Arable_Land_ha}$ (e.g. $\mu_{tobacco} = 0.35$)

4.2. Real Input Cost Ledger Calculations

For each selected crop, production outlays are itemized across six standard operational categories:

$$\text{Cost}_i = \text{Seed}_i + \text{BasalFert}_i + \text{TopDress}_i + \text{Labour}_i + \text{Herbicide}_i + \text{Transport}_i \quad (\text{MWK / ha})$$

$$\text{Total_Farm_Outlay} = \sum (x_i \cdot \text{Cost}_i)$$

$$\text{Budget_Surplus_or_Deficit} = \text{Budget} - \text{Total_Farm_Outlay}$$

Affordability Verdict	Mathematical Threshold	Advisory & System Action
Fully Funded	$\text{Surplus} \geq 0.15 \times \text{Budget}$	Optimal liquidity. Farmer maintains ≥15% cash contingency reserve for unexpected price movements.
Tight Liquidity	$0 \leq \text{Surplus} < 0.15 \times \text{Budget}$	Approved plan with zero contingency. Recommended to lock in fertilizer purchases early before retail inflation.
Funding Gap	$\text{Surplus} < 0$ (Deficit)	Plan infeasible. Interactive UI flags deficit amount and recommends reducing high-cost crops or substituting legumes.

4.3. Seasonal Risk Analysis Engine (`/scenarios/`)

The system stress-tests the approved farm plan against historical Malawian shocks:

Stress Scenario	Primary Drivers	Agronomic / Financial Perturbation
Drought (El Niño Cycle)	Extended 21-day dry spell during flowering.	Maize yield \(-45\%\), Legumes \(-25\%\), Cassava \(-10\%\). Revenue drops severely.
Fertilizer Inflation Shock	Forex scarcity, global natural gas spikes.	Commercial NPK & Urea prices \(+60\%\). Profit margins compress across high-input crops.
Commodity Price Crash	Regional market glut or bumper harvest.	Wholesale farm-gate prices \(-35\%\). Tests break-even thresholds.
Fall Armyworm Epidemic	Late rain onset followed by pest surge.	Maize yield \(-30\%\), chemical protection costs \(+40\%\).
Compound Shocks	Concurrent dry spell and input inflation.	Simultaneous \(-40\%\) yield shock and \(+50\%\) input price increase.

5. Real-Time Data Pipelines & Ingestion Protocols

5.1. WFP Malawi / Humanitarian Data Exchange (HDX) Pipeline

The system synchronizes commercial input prices directly from the public HDX/WFP price repository, requiring zero authentication tokens and operating resiliently:

- **Dataset Endpoint:** `https://data.humdata.org/api/3/action/package_show?id=wfp-food-prices-for-malawi`
- **Ingested Commodities:** Fertilizer (CAN), Fertilizer (urea), Fertilizer (D compound), Fertilizer (NPK), Maize (seed).
- **Conversion Protocol:**

  > `\( ext{Price per Hectare} = ext{Price per 50kg bag} imes \left( rac{ ext{Agronomic Application Rate (kg/ha)}}{50 ext{ kg}} ight)\)`

- **Execution Interface:** `python manage.py sync_wfp_prices [--dry-run] [--season 2025]`

5.2. DCCMS Weather & Agro-Advisory Pipeline

- Ingests daily precipitation, minimum/maximum temperature, and relative humidity from Chitedze, Kasungu, Lunzu, Mzuzu, and Salima stations.
- Calculates **Rainfall Onset**: \(\ge 25 ext{ mm}\) rainfall accumulated within 72 hours after November 1st without an ensuing 10-day dry spell.

6. UI Standards, Security & Architecture Governance

- **Professional Styling:** Strict prohibition of gamified copy or juvenile aesthetics. Employs formal institutional design: clean data grids, Lucide SVG iconography, and official agricultural terminology.
- **Localization:** Native bilingual architecture with Chichewa (*Chichewa cha alimi*) alongside English.
- **Zero Leakage Security:** API keys, production tokens, and infrastructure secrets decoupled into `.env` (strictly gitignored).
- **Test Integrity:** Complete continuous verification via Django test suite ensuring 79/79 passing unit, model, and view test assertions.